

App Dev Project Details

1.Student Details

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About Me: I am Student of IIT Madras BS degree program and Currently I am on the Diploma Level

2. Project Details

Project Title: Hospital Management System

Problem Statement:

To build a **Hospital Management System (HMS)** web application that allows **Admins, Doctors, and Patients** to interact with the system based on their roles.

Approach:

The app was built using Flask as the backend framework with a modular structure. The application is built as a Hospital Management System (HMS) with a role-based structure, allowing admins, doctors, and patients to interact efficiently. It supports managing appointments, medical records, and user activities, providing a streamlined and organized healthcare management experience.

3. AI/LLM Declaration

I used **ChatGPT (GPT-5)** to assist in writing SQLAlchemy model definitions, for making some part of the page responsive i used javascript in one place and little bit for CSS also

The extent of AI/LLM usage is around **5%** limited to **code suggestions and documentation formatting**.

All final implementation logic, debugging, and integration were done manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
SQLite	Lightweight local database for storing user data

5. Database Schema / ER Diagram

Tables:

Users — (id, department_id, username, email, password, role, phone, blacklisted, drexperience, drqualification, created_at)

departments — (id, name, description, blacklisted)

appointments — (id, patient_id, doctor_id, appointment_date, appointment_time, reason, status, diagnosis, prescription)

Treatments — (id, patient_id, appointment_id, visit_date, visit_number, visit_type, prescription, diagnosis_text, medicines)

doctor_availability — (id, doctor_id, date, morning, evening)

Relationships:

departments → **users** : one-to-many

users (patient) → **appointments** : one-to-many

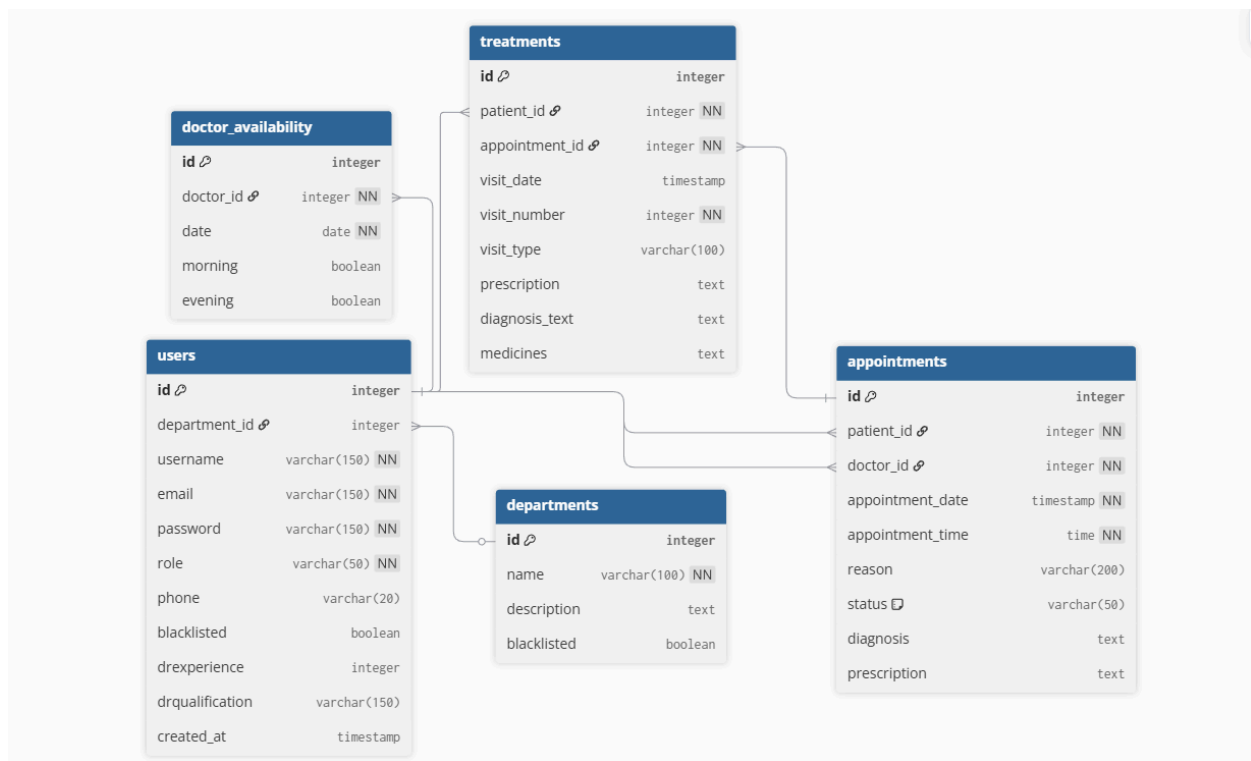
users (doctor) → **appointments** : one-to-many

appointments → **treatments** : one-to-one

users (patient) → **treatments** : one-to-many

users (doctor) → **doctor_availability** : one-to-many

ER Diagram:



Authentication

Endpoint	Method	Description
/register	GET, POST	Register a new patient
/login	GET, POST	Authenticate user and create session
/logout	GET	Logout user and clear session

Admin Dashboard

Endpoint	Method	Description
/admin	GET	Admin dashboard with search/filter functionality
/admin_dashboard/patient_history_by_admin/<patient_id>	GET	View patient history by admin
/admin_dashboard/add_department	GET, POST	Add a new department
/admin_dashboard/add_doctor	GET, POST	Add a new doctor
/admin_dashboard/edit_doctor/<doctor_id>	GET, POST	Edit doctor details
/admin_dashboard/edit_patient/<patient_id>	GET, POST	Edit patient details
/admin_dashboard/delete_doctor/<doctor_id>	POST	Delete a doctor
/admin_dashboard/delete_patient/<patient_id>	POST	Delete a patient

Endpoint	Method	Description
/admin_dashboard/blacklist_doctor/<doctor_id>	POST	Blacklist a doctor
/admin_dashboard/blacklist_patient/<patient_id>	POST	Blacklist a patient
/admin_dashboard/view_doctors_by_department/<dept_id>/doctors	GET	View doctors in a department
/admin_dashboard/delete_department/<dept_id>	POST	Delete a department
/admin_dashboard/blacklist_department/<dept_id>	POST	Blacklist a department

Doctors Dashboard

Endpoint	Method	Description
/doctor_dashboard	GET	Doctor dashboard with upcoming and completed appointments
/doctor/availability	GET, POST	Set or update doctor availability

/patient_dashboard/mark_as_complete_by_dr/<appointment_id>	GET	Mark appointment as complete
/doctor_dashboard/add_treatment/<appointment_id>	GET, POST	Add treatment record
/patient_dashboard/cancel_treatment/<appointment_id>	GET	Cancel appointment (doctor)
/doctor_dashboard/view_patient_history/<patient_id>	GET	View patient treatment history

Patient Dashboard

Endpoint	Method	Description
/patient_dashboard	GET	Patient dashboard with doctors, appointments, and treatments
/patient_dashboard/view_doctors_by_patient/<dept_id>/doctors	GET	View doctors by department
/patient_dashboard/view_doctors_by_patient/doctor_detail_by_patient/<doctor_id>	GET	View doctor details

/patient_dashboard/cancel_appt_by_pt/<appointment_id>	POST	Cancel appointment
/view_doctor_by_patient/check_avail_by_patient/<doctor_id>	GET, POST	Check doctor availability
/check_avail_by_patient/booking_appointment/<doctor_id>	POST	Book appointment
/patient_dashboard/edit_profile_by_patient	GET, POST	Edit patient profile
/patient_dashboard/history_patient	GET	View patient appointment & treatment history

Video Presentation:

<https://drive.google.com/drive/folders/1Ing2QBXhWWfc-yBGUCsC1rID-RyJ7i0W>